

Solar O appendix

Sun · O · downloadable appendix

Solar oxygen: 8.660 ± 0.101

Asplund, Amarsi & Grevesse 2021: 8.69 ± 0.04 · this measurement -0.030 dex

Kitt Peak · red-optical · Reference Grade · SET-AGSS21 · ENGINE-A-3DNLTE · n = 6. The top-ranked admitted product (Reference Grade, then the most complete treatment, then line count, then total uncertainty); independent products and engines are not averaged together.

Highlighted Products

VIS

8.934 ± 0.395

± 0.309 stat · ± 0.246 syst

FORB-OI_6300 · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied · solar_harps_molecfits_corrected · Codex Grade

red-optical

8.660 ± 0.101

± 0.044 stat · ± 0.043 syst

SET-AGSS21 · O I · Synth · 3D-NLTE · Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not supplied · solar_kpno_kurucz2005_corrected · Reference Grade

NIR

8.784 ± 0.121

± 0.053 stat · ± 0.173 syst

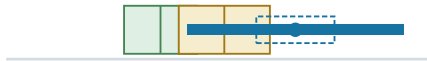
SET-AGSS21 · O I · Synth · 3D-NLTE · Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not supplied · solar_iag · Reference Grade

Error-bar Forest Plot

VIS — harps solar_harps_molecfits_corrected A(O) dex; axis 8.60-9.27		
FORB-OI_6300 · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Codex Grade · ALL · n=1		8.934 ± 0.309 stat ± 0.246 syst
VIS axis: 8.60 · 8.76 · 8.93 · 9.10 · 9.27 A(O) dex		
red-optical — iag_fts_solar_atlas solar_iag A(O) dex; axis 8.52-8.99		
SET-AGSS21 · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Reference Grade · ALL · n=6		8.820 ± 0.050 stat ± 0.043 syst
SET-AGSS21 · O I · Synth · 3D-NLTE · Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not supplied Reference Grade · ALL · n=6		8.659 ± 0.045 stat ± 0.043 syst
SET-AGSS21 · O I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied Reference Grade · ALL · n=6		8.689 ± 0.040 stat ± 0.043 syst

red-optical — iag_fts_solar_atlas solar_iag A(O) dex; axis 8.52-8.99

indicator not supplied · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied
Codex Grade · ALL · n=2



8.838
±0.119 stat ±0.043 syst

indicator not supplied · O I · Synth · 3D-NLTE · Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not supplied
Codex Grade · ALL · n=2



8.677
±0.108 stat ±0.043 syst

indicator not supplied · O I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied
Codex Grade · ALL · n=2



8.706
±0.095 stat ±0.043 syst

red-optical — kpno_solar_atlas solar_kpno_kurucz2005_corrected A(O) dex; axis 8.52-8.99

SET-AGSS21 · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied
Reference Grade · ALL · n=6



8.820
±0.048 stat ±0.043 syst

SET-AGSS21 · O I · Synth · 3D-NLTE · Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not supplied
Reference Grade · ALL · n=6



8.660
±0.044 stat ±0.043 syst

SET-AGSS21 · O I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=6



8.690
±0.038 stat ±0.043 syst

indicator not supplied · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied
Codex Grade · ALL · n=2



8.841
±0.122 stat ±0.043 syst

indicator not supplied · O I · Synth · 3D-NLTE · Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not supplied
Codex Grade · ALL · n=2



8.680
±0.112 stat ±0.043 syst

indicator not supplied · O I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied
Codex Grade · ALL · n=2



8.709
±0.098 stat ±0.043 syst

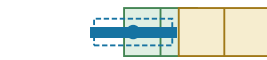
red-optical — kpno_solar_atlas solar_kpno_molecfit_corrected A(O) dex; axis 8.52-8.99

SET-AGSS21 · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied
Reference Grade · ALL · n=6



8.820
±0.052 stat ±0.043 syst

SET-AGSS21 · O I · Synth · 3D-NLTE · Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not supplied
Reference Grade · ALL · n=6



8.660
±0.047 stat ±0.043 syst

SET-AGSS21 · O I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=6



8.690
±0.041 stat ±0.043 syst

indicator not supplied · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied
Codex Grade · ALL · n=2



8.833
±0.132 stat ±0.043 syst

indicator not supplied · O I · Synth · 3D-NLTE · Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not supplied
Codex Grade · ALL · n=2



8.672
±0.121 stat ±0.043 syst

red-optical — kpno_solar_atlas solar_kpno_molecfits_corrected A(O) dex; axis 8.52-8.99

indicator not supplied · O I · Synth · 1D-NLTE ·
Bergemann · SYNTH · ENGINE-A · line set not
supplied
Codex Grade · ALL · n=2

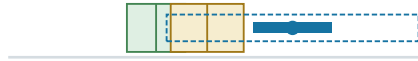


8.701
±0.107 stat ±0.043 syst

red-optical axis: 8.52 · 8.64 · 8.76 · 8.88 · 8.99 A(O) dex

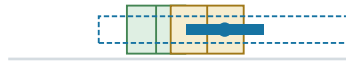
NIR — iag_fts_solar_atlas solar_iag A(O) dex; axis 8.49-9.08

SET-AGSS21 · O I · Synth · 1D-LTE · SYNTH · 1D-LTE ·
line set not supplied
Reference Grade · ALL · n=9



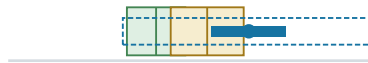
8.877
±0.054 stat ±0.173 syst

SET-AGSS21 · O I · Synth · 3D-NLTE · Amarsi · SYNTH
· ENGINE-A-3DNLTE · line set not supplied
Reference Grade · ALL · n=9



8.784
±0.053 stat ±0.173 syst

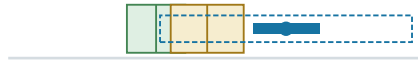
SET-AGSS21 · O I · Synth · 1D-NLTE · Bergemann ·
SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=9



8.817
±0.051 stat ±0.173 syst

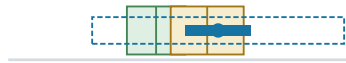
NIR — kpno_solar_atlas solar_kpno_kurucz2005_corrected A(O) dex; axis 8.49-9.08

SET-AGSS21 · O I · Synth · 1D-LTE · SYNTH · 1D-LTE ·
line set not supplied
Reference Grade · ALL · n=9



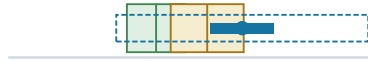
8.868
±0.046 stat ±0.173 syst

SET-AGSS21 · O I · Synth · 3D-NLTE · Amarsi · SYNTH
· ENGINE-A-3DNLTE · line set not supplied
Reference Grade · ALL · n=9



8.775
±0.045 stat ±0.173 syst

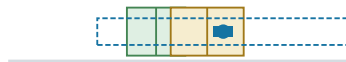
SET-AGSS21 · O I · Synth · 1D-NLTE · Bergemann ·
SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=9



8.808
±0.043 stat ±0.173 syst

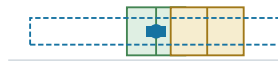
NIR — kpno_solar_atlas solar_kpno_molecfits_corrected A(O) dex; axis 8.49-9.08

SET-AGSS21 · O I · Synth · 1D-LTE · SYNTH · 1D-LTE ·
line set not supplied
Reference Grade · ALL · n=9



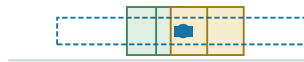
8.782
±0.013 stat ±0.173 syst

SET-AGSS21 · O I · Synth · 3D-NLTE · Amarsi · SYNTH
· ENGINE-A-3DNLTE · line set not supplied
Reference Grade · ALL · n=9



8.690
±0.013 stat ±0.173 syst

SET-AGSS21 · O I · Synth · 1D-NLTE · Bergemann ·
SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=9



8.727
±0.013 stat ±0.173 syst

NIR axis: 8.49 · 8.64 · 8.78 · 8.93 · 9.08 A(O) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Solid bars show the supplied statistical uncertainty; wireframes show the supplied systematic uncertainty. The statistical basis and any incomplete error budget remain attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every oxygen value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the oxygen abundance as the free parameter. The synthesis includes the atomic lines and the molecular bands (CN, CH, C₂, OH and others) in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, blends and coupling to the other C/N/O abundances, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

Solar oxygen context

Oxygen is traced by forbidden [O I], permitted O I and molecular OH indicators where available. Blends, departures from LTE and atmospheric structure affect these families differently, so their results retain separate identities.

[C/O indicator strategy](#) · [Methodology](#)

Open provenance decisions

O I 615.8nm: □ DISAGREES — reported to Ryan, not resolved. . Store log gf -0.409 (); summed pull -0.2966 (B+).

O I 630.0nm: □ DISAGREES — reported to Ryan, not resolved. □ store claims A, pull publishes B+. Store log gf -9.717 (A); summed pull -9.7744 (B+).

OPEN. Tachiev and NIST differ systematically by -0.016 dex on the 777 triplet and -0.005 dex on 8446. AGSS21 names Tachiev for N i only and states no O i gf source. Both values are reported in oi_tachiev_vs_nist.csv; NIST is what the store already carried and this script does not change it. Ryan decides.

[Adjudication record](#) · [Indicator gf checks](#)

Download the evidence

[Download PDF](#) — Solar O appendix

[Source feed](#) — includes quarantined fit outputs, not adopted abundances

[Complete product visibility audit](#)

[Shared website/PDF report](#)

[Reproducibility manifest](#)

[Complete references and product-to-source bindings](#)

Reproducibility

Generated from the pinned CNO campaign snapshot. Source commit
7ee783c2c04584338b0dbd82e0183ffc56b32e07

Feed O.json v1.84 · updated 2026-10-09T08:06:39Z

Generated 2026-10-09T02:09:29-06:00 (source commit timestamp)

Feed SHA-256 8367aa493f0afa85cbb11fedd4a85c2146430bad869c9005ca46cc6cc72aad6c

Generator scripts/generate_cno_publication.py v1.0.0

[Pinned scientific source](#)

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar O atomic/molecular indicator lineage and atmospheric sensitivity; comparison literature, not an adopted Codex abundance. Sources: [\[1\]](#), [\[2\]](#).

Atomic data and gf authorities. The recorded O I/O II base pedigree is Wiese-Fuhr-Deters / Opacity Project theory. Per-transition exceptions and the open O I source decisions remain explicit. Sources: [\[3\]](#), [\[4\]](#), [\[5\]](#).

Atomic data and gf authorities. O I triplet source record: NIST (this script) — Tachiev NOT adopted; OPEN — referred to Ryan (RYA-1214 EOS) Sources: [\[3\]](#), [\[6\]](#).

Model and grid sources. Registered O I line/grid lineage: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — line-by-line 3D-NLTE grids built on the Amarsi+2018 O I model atom. The current feed treatment remains authoritative; a grid reference does not imply a correction was applied. Sources: [\[7\]](#), [\[8\]](#).

Atomic data and gf authorities. Atomic O I entry; source-family pedigree only, not a substitute for a per-transition gf join. Sources: [\[3\]](#), [\[4\]](#), [\[5\]](#).

Model and grid sources. Feed treatment: 1D-LTE; no NLTE/3D correction is claimed for this entry. Turbospectrum is the synthesis implementation. Sources: [\[9\]](#).

Model and grid sources. Registered C/O model lineage; per-product grid provenance must confirm the applied treatment. Run provenance: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — 3D non-LTE - 1D LTE on STAGGER (table2 C I / table3 O I); PER-LINE additive correction, grid lines only (match ≤ 0.05 Å) Sources: [\[7\]](#), [\[8\]](#).

Model and grid sources. Registered C/O model lineage; per-product grid provenance must confirm the applied treatment. Run provenance: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — 1D non-LTE - 1D LTE (table5 C I / table6 O I); PER-LINE additive correction, grid lines only (match ≤ 0.05 Å) Sources: [\[7\]](#), [\[8\]](#).

Special diagnostics. The synthesis sidecar names Johansson's Ni I blend data, Storey & Zeippen's forbidden-line gf lineage and Caffau's CO5BOLD comparison anchor. The feed entry remains 1D-LTE and quarantined; the sidecar's cited anchor is not promoted to a measured abundance. Sources: [\[10\]](#), [\[11\]](#), [\[12\]](#).

Product-to-source index

O-products-2f26fcc4de61 · products · NIR · solar_iag · SET-AGSS21 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

O-products-ee093b5cd43b · products · NIR · solar_iag · SET-AGSS21 · ENGINE-A-3DNLTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-dcccc1bf145a · products · NIR · solar_iag · SET-AGSS21 · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-7570ea595567 · products · NIR · solar_kpno_kurucz2005_corrected · SET-AGSS21 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

O-products-34b4e514053b · products · NIR · solar_kpno_kurucz2005_corrected · SET-AGSS21 · ENGINE-A-3DNLTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-f76024e80aca · products · NIR · solar_kpno_kurucz2005_corrected · SET-AGSS21 · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-97dff9c24866 · products · NIR · solar_kpno_molecfit_corrected · SET-AGSS21 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

O-products-4ce6582718b9 · products · NIR · solar_kpno_molecfit_corrected · SET-AGSS21 · ENGINE-A-3DNLTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-dbd116adb0fe · products · NIR · solar_kpno_molecfite_corrected · SET-AGSS21 · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-951619c74bb3 · products · VIS · solar_harps_molecfite_corrected · FORB-OI_6300 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#), [\[10\]](#), [\[11\]](#), [\[12\]](#)

O-products-400d2fb23109 · products · red-optical · solar_iag · SET-AGSS21 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

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O-products-14585017d015 · products · red-optical · solar_iag · SET-AGSS21 · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

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O-products-5dd7a576c91e · products · red-optical · solar_kpno_kurucz2005_corrected · not supplied · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

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O-products-192c0fab28e1 · products · red-optical · solar_kpno_molecfite_corrected · SET-AGSS21 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

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O-products-6fccf43002bd · products · red-optical · solar_kpno_molecfite_corrected · SET-AGSS21 · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

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O-products-d115064cbf77 · products · red-optical · solar_kpno_molecfite_corrected · not supplied · ENGINE-A-3DNLTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-0d45f4f81b47 · products · red-optical · solar_kpno_molecfite_corrected · not supplied · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

Complete applicable bibliography

Solar abundance and indicator analyses

[1] — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. *A&A* 653, A141 DOI: 10.1051/0004-6361/202140445. [source record](#)

[2] — Amarsi et al. 2021, *A&A* 656 A113 DOI: 10.1051/0004-6361/202141384. [source record](#)

Atomic data and gf authorities

[3] — Kramida, A.; Ralchenko, Yu.; Reader, J.; NIST ASD Team (2024). NIST Atomic Spectra Database (version 5.11/5.12). National Institute of Standards and Technology, Gaithersburg, MD DOI: 10.18434/T4W30F. [source record](#)

[4] — Wiese, W. L.; Fuhr, J. R. (2006). New Critical Compilations of Atomic Transition Probabilities for Neutral and Singly Ionized Carbon, Nitrogen, and Iron. Proceedings of the NASA Laboratory Astrophysics Workshop (14-16 February 2006, UNLV, Las Vegas) [source record](#)

[5] — Wiese, W. L.; Fuhr, J. R.; Deters, T. M. (1996). Atomic Transition Probabilities of Carbon, Nitrogen, and Oxygen: A Critical Data Compilation. JPCRD Monograph 7. [source record](#)

[6] — Tachiev & Froese Fischer 2002, A&A 385, 716 DOI: 10.1051/0004-6361/20011816. [source record](#)

Model and grid sources

[7] — Amarsi, A. M.; Nissen, P. E.; Skuladottir, A. (2019). Carbon, oxygen, and iron abundances in disk and halo stars. A&A 630, A104 DOI: 10.1051/0004-6361/201936265. [source record](#)

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[9] — Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum. A&A 669, A43 DOI: 10.1051/0004-6361/202243673. [source record](#)

Special diagnostics

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